

Description

V-rings are rotary seals that can perform numerous jobs in their function of sealing rotating shafts:

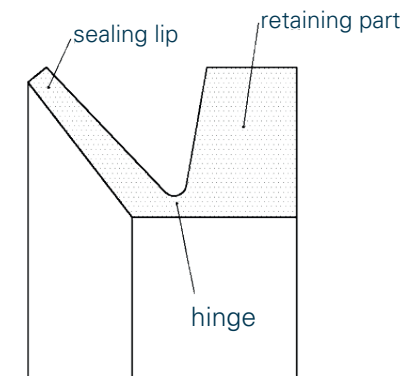
- sealing against the penetration of dirt, dust, water or watery pollutants,
- combination with other sealing elements, e.g. the protection of a radial shaft sealing ring against external contamination, or
- sealing against grease leaking from a housing.

Advantages of V-rings

- diverse fields of application
- no high requirements as regards the shaft and housing designs
- low friction
- high circumferential speeds are possible as a result of reduced friction
- relatively insusceptible to coaxiality and radial eccentricity
- long service life
- easy installation

Design and function

V-rings are made entirely of elastomer material. Their V-shaped profile consists of a relatively solid retaining part and a flexible sealing lip connected by a „hinge“.

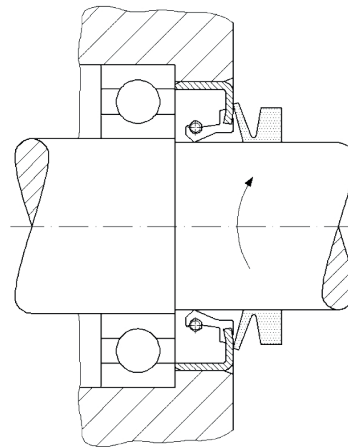


Profile of a V-ring, type VR-A10

V-rings are expanded during assembly, pushed to their intended position on the shaft, and hold onto the shaft surface by their internal tension.

V-rings will rotate with the shaft and provide sealing in axial direction against a counter face that is positioned vertically to it. As a result of the diverse fields of application for V-rings the counter face can be formed by

- the housing itself,
- a housing cover,
- a pressed-in metal sheet,
- the face of a roller bearing's external ring,
- the metallic stiffening ring of a radial shaft seal
- etc.



Protection of a radial shaft seal against external contamination

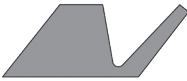
When not in motion, the sealing lips adhere to the counter face due to the designed preload. Once the shaft starts rotating, the centrifugal force acts in radial direction on the sealing lip, while the contact pressure drops with the increasing circumferential speed.

The sealing lip starts to lift off the counter face from a circumferential speed of approx. 15-20 m/s and the contact pressure drops to almost zero. The friction loss will increase in a linear fashion up to approx. 10m/s. After that time, as the speed reaches toward circumferential speed of 15-20 m/s the contact pressure will drop to near zero.

A V-ring's sealing effect is based on the contact between sealing lip and counter face and, additionally, on the centrifugal effect caused by the V-ring's rotation. Once the sealing lip takes off at a very high circumferential speed, the V-ring will act as gap ring and centrifugal disk.

Construction types/designs

In order to cover the wide range of possible applications, we have 3 standard designs in stock, made of 2 different materials each (NBR and FKM).

Profilskizze	Bauform	Dimensional range in stock [mm]	Remarks/comments
	VR-A10	3 - 1000 >1000 on request	most commonly used standard design with straight sealing back
	VR-S10	5 - 199	the same design as VR-A10 in the lip area, complemented by a conically extended retaining part, thus providing a better seat on the shaft
	VR-L10	110 - 600 >600 on request	small profile geometry for compact installation conditions uniform profile for all diameters

We can offer further construction types/designs on request, e.g.

VR-E10

VR-AX10